8D PROBLEM SOLVING REPORT (QUALITY + SAFETY + EMS)

Day - 3

Project Reference Code:	`AER-8D2026-TB03`
Target Industry:	Aerospace Precision Manufacturing
Component Name:	High Pressure Gas Turbine Blade (Inconel 718)
Manufacturing Process:	5-Axis CNC Precision Grinding
Applicable Standards:	AS9100 Rev D, ISO 9001, ISO 14000 (series), ISO 45001, ISO 19011
Inspection Criteria:	Fluorescent Penetrant Inspection (FPI) & CMM Measurement
Target Specification (Thickness):	12.500 ± 0.015 mm (Max: 12.515 mm | Min: 12.485 mm) *Acc. to Taguchi: 12.500 mm (Better) — (Taguchi Philosophy used)*
Defect Identified:	Thermal Micro-cracks (Grit Burn) & Dimensional Thickness Drift
Baseline Rejection Rate:	6.8% → 5.4%
Date of Report:	30 July, 2026

D1: Team Formation (CFT Formation)

1. **Manufacturing Department**
2. **Production Department**
3. **Operational Department**
4. **Quality Management Department & (QA / QC / QI)**
5. **Supply Chain & Procurement Department**
6. **Management Department** *(For overall balance & management of all departments)*
7. **Research & Development Department** *(For making methodologies based on overall research)*
8. **Finance & Accounting Department** *(For management of overall data & metrics system and its analysis — Responsible for financial budget for overall project analysis.)*
9. **Human Resources Department** *(For communication & maintaining connection b/w team — Responsibility of maintaining ethical & strategic workspace planning.)*
10. **Legal and Compliance Department** *(For maintaining government regulations)*
11. **Maintenance / Facility Management:** *(For servicing of tools & machining)*
12. **IT Department:** *(For maintaining data security)*

D2: Problem Description (5W2H Framework)

What:

1. Found thermal micro-cracks on FPI.
2. Dimensional variation in CMM Inspections — Blade root thickness exceeds limit (12.500 ± 0.015 mm).

> **Note:** Rejection rate 6.8% → 5.4%
> During 5-Axis CNC grinding, due to more vibration, sound level reaches above 95 dB (which is harmful for worker, violates ISO 45001 Worker Safety).

Where:

Aerospace precision machining cell & 5-Axis CNC grinding line.

When:

Batch Inspection During Batch inspection for (TB-2026.07) (30 July 2026).

Who:

- Overall system
- Delay in delivery
- Effect on customer retention / requirements
- Assembly Plant
- Effect on worker
- Effect on Environment
- Aerospace Engine Quality (AS9100)

Why?

1. Because in Aerospace industry **Zero Defect** philosophy of Philip Crosby is followed with integration of ISO 9001 + AS9100.
2. As per Taguchi specification, must be near to target to prevent defect.
3. Effecting overall Quality & violating standard norms for (QMS, EMS & Worker Safety).
4. Increasing Internal and External failure cost and minimizing prevention cost.

How Much?

Rejection rate reaches to 5.4%. Due to which rework + scrap both cost increases (COPQ), and C_{pk} index is decreasing.

D3: Containment Actions & Measurement Validation

1. Immediate Containment

- Isolation of stock for Batch (TB-2026-07).
- Putting red tag on all parts (ISO 9001:2015, Clause 8.7).
- Sorting & Re-Inspectioning of all On Parts with help of FPI & CMM.
- Stopping production line at station of 5-Axis CNC Grinding.

2. Measurement System Analysis (MSA)

- **Gauge R&R check of CMM:** Due to which GR&R = 6.8% < 10% (CMM error is correct).
- **Calibration of FPI System** for prevention of false signal & response: NDT → Micro-X-ray, UV Testing

D4: Root Cause Analysis (RCA)

Tools Used: Fishbone (Ishikawa), 5-Why, FMEA

A) Fishbone (Ishikawa) Diagram

- **Man:**
  - Unskilled & level of training for specific controlling of machine equipment.
  - Due to noise levels, loss of concentration (violating ISO 45001).
  - Tiredness due to noise (strain).
- **Machine:**
  - Vibration, noise at 5-Axis CNC grinding.
- **Material:**
  - Level of tool wear resistant material.
  - Lack of noise prevention and vibration absorbing material like glass wool etc.
  - Inconel 718 → Thermal sensitive and coolant is unable to reduce thermal stress at high temperature.
- **Method:**
  - Level of high-end technological method used for early detection at FPI & CMM and at 5-Axis CNC grinding.
  - Due to unfixing of dressing interval which creates clogging of wheel due to friction generating.
- **Measurement:**
  - MSA (GR&R = 6.8% < 10%) → CMM Good (✓).
  - No root cause found in measurement.
  - *Note:* But to prevent failure, always follow MSA + Zero Defect philosophy + Kaizen. Use 6-Sigma → for variation checking at CMM.
- **Environment:**
  - Due to vibration & high speed of grinding, noise creation.

B) 5-Why Analysis

1. **Why Micro-cracks & Dimension variation?**
   → Due to heat & vibration. *(Apply Process FMEA / Implement Kaizen)*
2. **Why Excessive local heat & vibration?**
   → Due to clogging of surface and machine is unable to absorb vibration. *(Apply PFMEA)*
3. **Why Grinding wheel clogging & Vibration?**
   → Due to friction. *(Apply Design FMEA)*
4. **Why No Automated Dressing & Damping?**
   → Lack of In-Situ sensors.
5. **Root Cause:**
   [Thermal / Tool wear Monitoring In-Situ sensors are not used] & Absence of vibration absorbing & damping equipment and material.

C) FMEA

FMEA: PFMEA, DFMEA, SFMEA

Assign (S), (O), (D) for each analysis and calculate RPN No.

For better prevention apply: Six-Sigma → DMAIC + PFMEA

D5: Permanent Corrective Actions (PCA)

1. For preventing micro-cracks, use **Advanced 3D printing technology**.
2. Use of NDT methods like **UV Testing, Micro-X-ray** for checking of defects.
3. Apply **Poka-Yoke** methodology like (AI Integrated Poka-Yoke automated sensors).
4. Integration of **In-situ sensors**. *(ISO 9001:2015, AS9100)*
5. **For safety of worker (ISO 45001) & vibration absorption:**
    - Viscoelastic Damping pads
    - Acoustic Enclosure (Noise reduction) → *(Due to which noise level intensity decreased)*
6. **For Environment:**
    - Apply EMS Standard → ISO 14000
    - ISO 14001, ISO 14004, ISO 140011, ISO 140012, ISO 140013
    - *For PES:* ISO 14021, ISO 14022, ISO 14023

D6: Validation & Verification Process

1. **SPC (Statistical Process Control):**
   Control Chart: S-chart (Deviation), R-chart (Range / Variation)
2. **C_{pk} index measured** which indicates control limit specification.
3. **Re-inspection** for FPI & CMM.
4. **Environmental Labelling & Auditing** Performed.

D7: Systematic Prevention

1. Use of (PDCA + Kaizen) to prevent failure.
2. Zero-defect system implementation (Philip Crosby).
3. Use of AI Integration + In-situ sensors.
4. Use of high-end NDT technology for micro-cracks and micro-pores.
5. Implementation of SQC.

D8: Closure

1. Signature of all departments head & manager.
2. **Employee Empowerment (Edward Deming):**
    - Employee appreciation (Edward Deming)
    - Providing of incentives & achievement trophies to employees for their hard work.
    - Provide scope of future or upcoming project.